

2. Market Microstructure and Crypto Market

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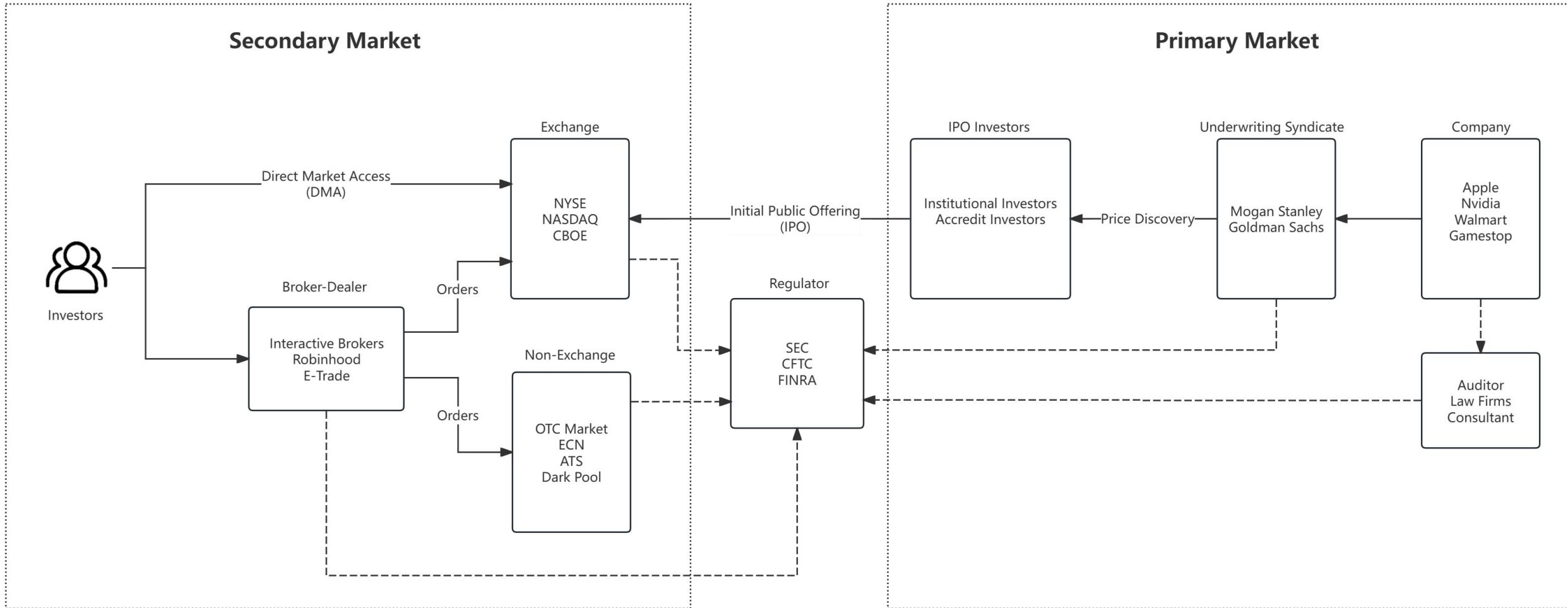
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1. Market Structure

1. Market Structure: Stock Market

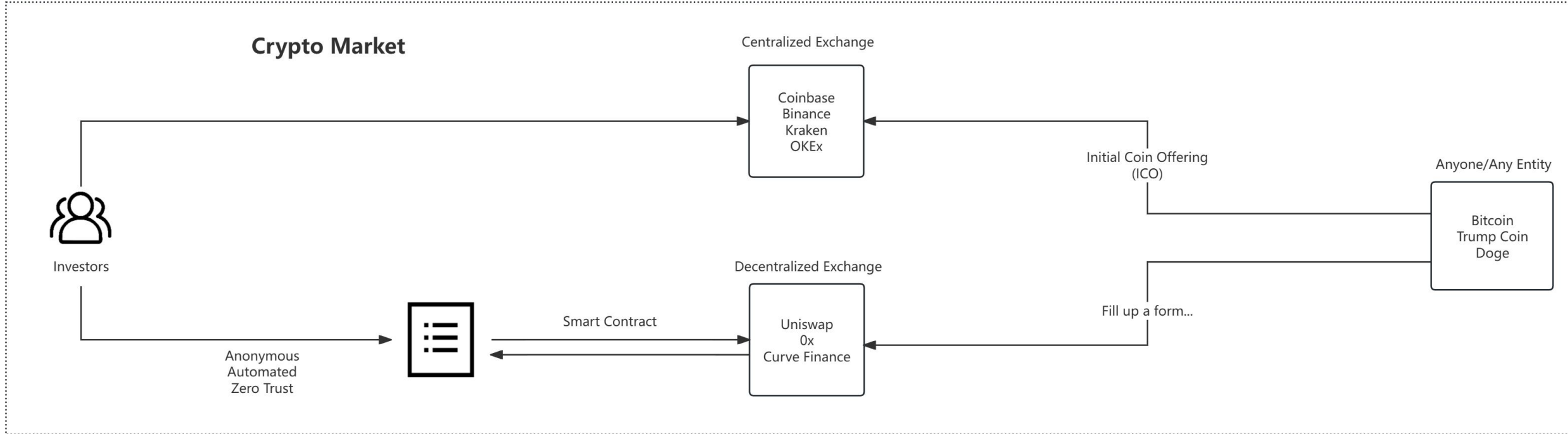


- All participants are subject to regulation.

1. Market Structure: IPO



1. Market Structure: Crypto Market



- The spot crypto market still lacks proper regulation.
- Most of centralized exchanges are now complied for anti-money laundry.

1. Market Structure: ICO

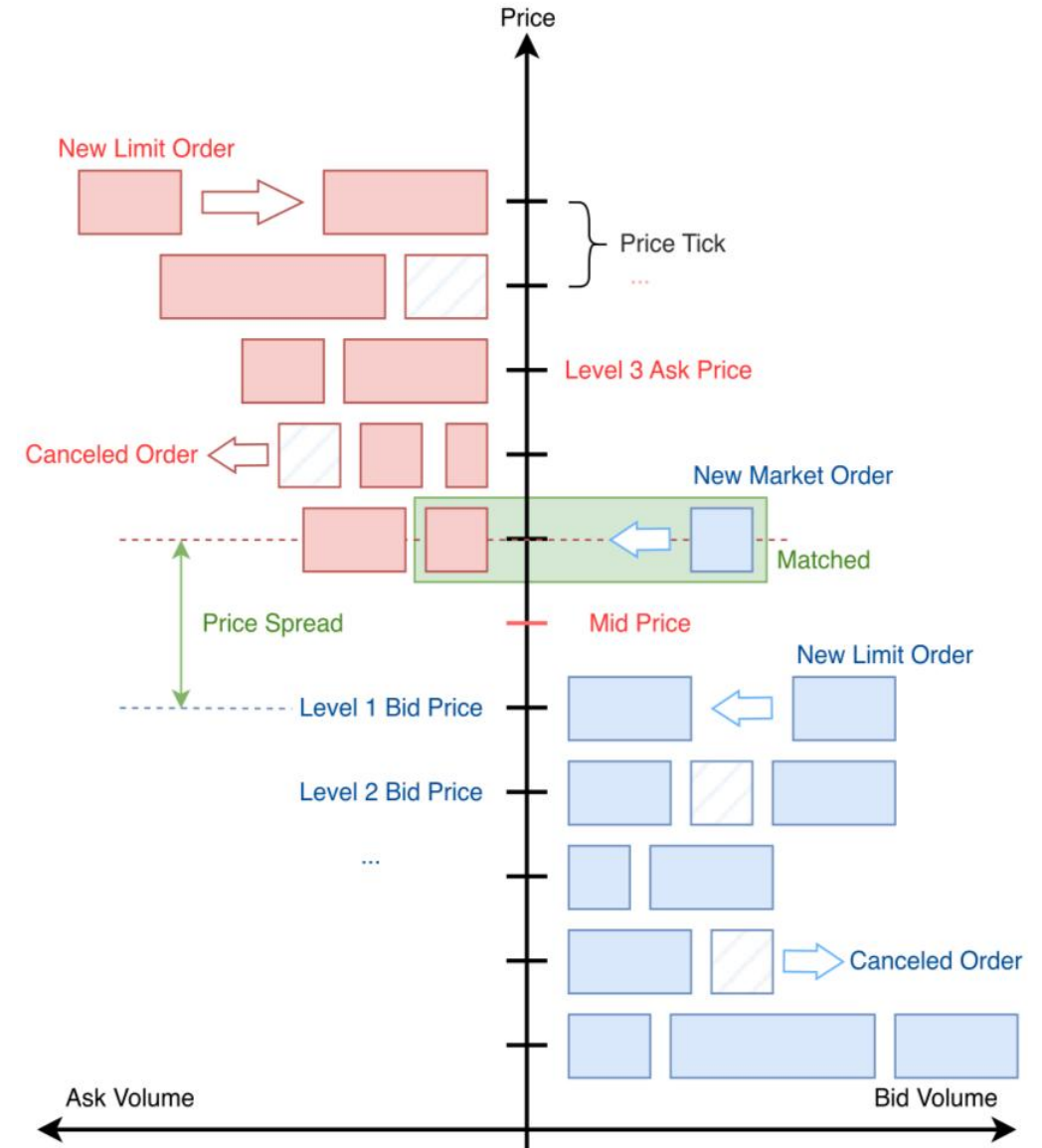


- During an ICO process, a CEX is usually playing multiple roles such as investor, regulator, auditor, consultant, promoter, etc.
- After a token is listed, a CEX is also playing roles as exchange, market maker, clearance house, custodian, investor.
- SEC sued Coinbase for operating as an Unregistered Securities Exchange, Broker, and Clearing Agency in 2023. However, the case has been dropped in 2025.

2. Market Microstructure

2. Market Microstructure: Limit Order Book (Double Auction)

- **Double Auction:** a process in which both buyers and sellers submit offers simultaneously.
- **Market Order:** an order to buy or sell a stock at the market's best available price. It typically ensures an execution but doesn't guarantee a specific price.
- **Limit Order:** an order to buy or sell a stock with a restriction on the maximum price to be paid (with a buy limit) or the minimum price to be received (with a sell limit). If the order is filled, it will only be at the specified limit price or better. However, there is no assurance of execution.
- **Price Priority > Time Priority**
- **FIFO (First-In, First-Out):** The order that arrived earliest (or has the oldest timestamp) will be filled first when multiple orders exist at the same price level.



2. Market Microstructure: Limit Order Book

Conditional orders

- Hidden parts of Icebergs
- Stop orders
- Order-sends-order chains
- Time-on-force, and more...



Invisible with any market data

New orders or triggered conditional orders enter the Orders Queue

Orders advance in the orders queue →

				Price	Total size
		20	15	1111	35
25	10	20		1110	55
20	10	30		1109	60
	20	20	10	1108	50

Visible with Order-by-Order or Market-by-Order (MBO) data

Visible with Market-by-Price (MBP) data

Ask Order Book:
Collection of resting
Sell Limit Orders

Best Bid & Ask

Bid Order Book:
Collection of resting
Buy Limit Orders

**Order Book
or
Market Depth**

many more price levels

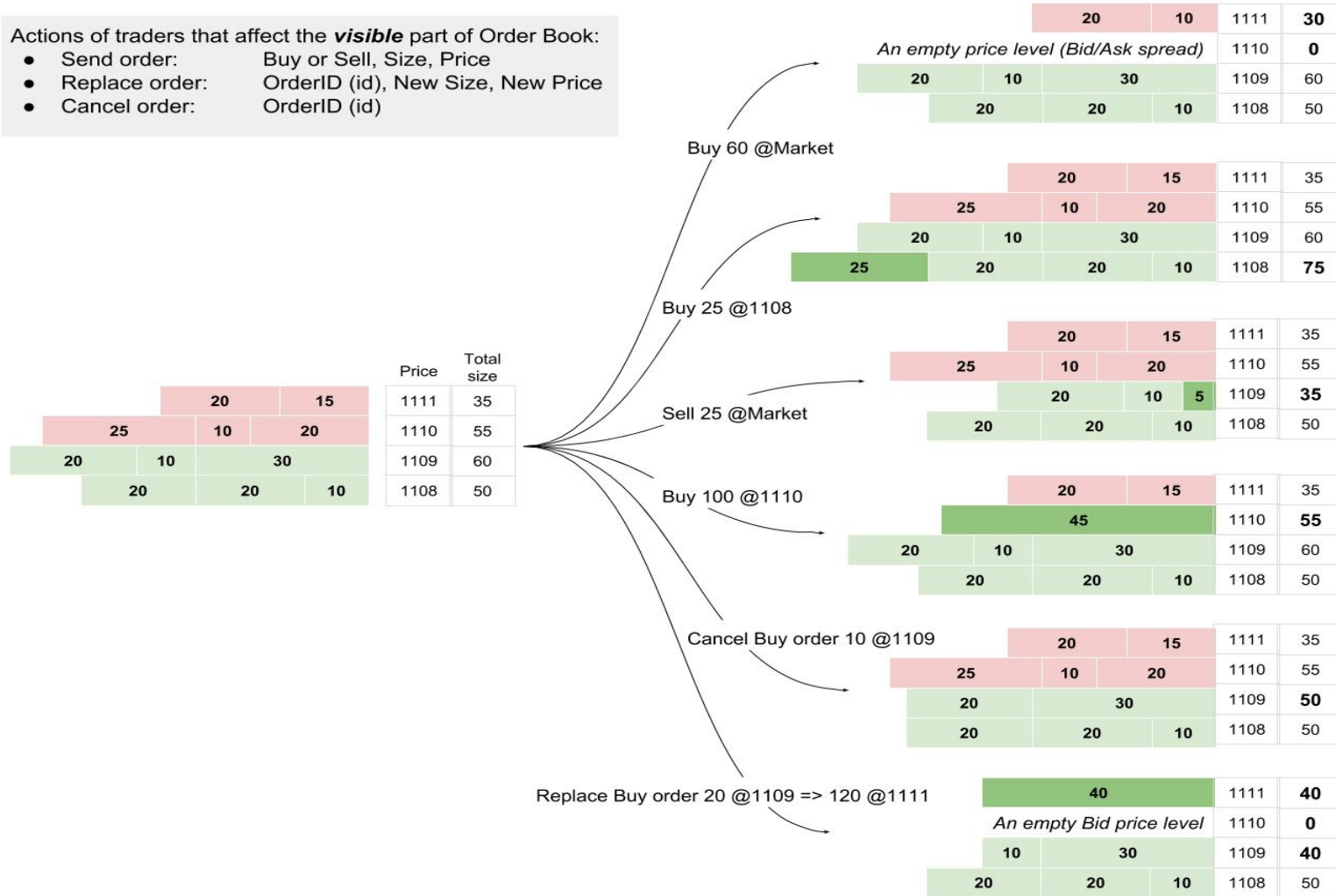
many more price levels

2. Market Microstructure: Limit Order Book

Illustration of Matching Engine processing actions of traders using Price-Time priority Matching Algorithm, aka FIFO.

Actions of traders that affect the **visible** part of Order Book:

- Send order: Buy or Sell, Size, Price
- Replace order: OrderID (id), New Size, New Price
- Cancel order: OrderID (id)



Corresponding market data updates (schematic only)			
MBP (e.g. CME)	MBO (e.g. CME)	MBO (e.g. Nasdaq)	SRC (patent pending)
Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Execution: 5 @1111 Set Ask 0 @1110 Set Ask 30 @1111	Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Execution: 5 @1111 Delete Sell 20 @1110 Delete Sell 10 @1110 Delete Sell 25 @1110 Update Sell 15 @1111 ->10 @1111	Executed: 20 @1110 Executed: 10 @1110 Executed: 25 @1110 Executed: 5 @1111	Send Buy 60 @Market
Set Bid 75 @1108	Add Buy 25 @1108	Add Buy 25 @1108	Send Buy 25 @1108
Execution: 25 @1109 Set Bid 35 @1109	Execution: 25 @1109 Update Buy 30 @1109 -> 5 @1109	Executed: 25 (of 30) @1111	Send Sell 25 @Market
Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Set Bid 45 @1110 Set Ask 0 @1110	Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Add Buy 45 @1110 Delete Sell 20 @1110 Delete Sell 10 @1110 Delete Sell 25 @1110	Executed: 20 @1110 Executed: 10 @1110 Executed: 25 @1110 Add Buy 45 @1110	Send Buy 100 @1110
Set Bid 50 @1109	Delete Buy 10 @1110	Delete Buy order 10 @1110	Cancel Buy 10 @1109
Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Execution: 15 @1111 Execution: 20 @1111 Set Bid 40 @1111 Set Bid 40 @1109 Set Ask 0 @1110 Set Ask 0 @1111	Execution: 20 @1110 Execution: 10 @1110 Execution: 25 @1110 Execution: 15 @1111 Execution: 20 @1111 Update Buy 20 @1109 -> Buy 40 @1111 Delete Sell 20 @1110 Delete Sell 10 @1110 Delete Sell 25 @1110 Delete Sell 15 @1111 Delete Sell 20 @1111	Executed: 20 @1110 Executed: 10 @1110 Executed: 25 @1110 Executed: 15 @1111 Executed: 20 @1111 Add Buy 40 @1111	Replace Buy 20@1109 -> 120 @1111

2. Market Microstructure: Limit Order Book (Levels of Order Flow Data)

Levels 1 and 2:

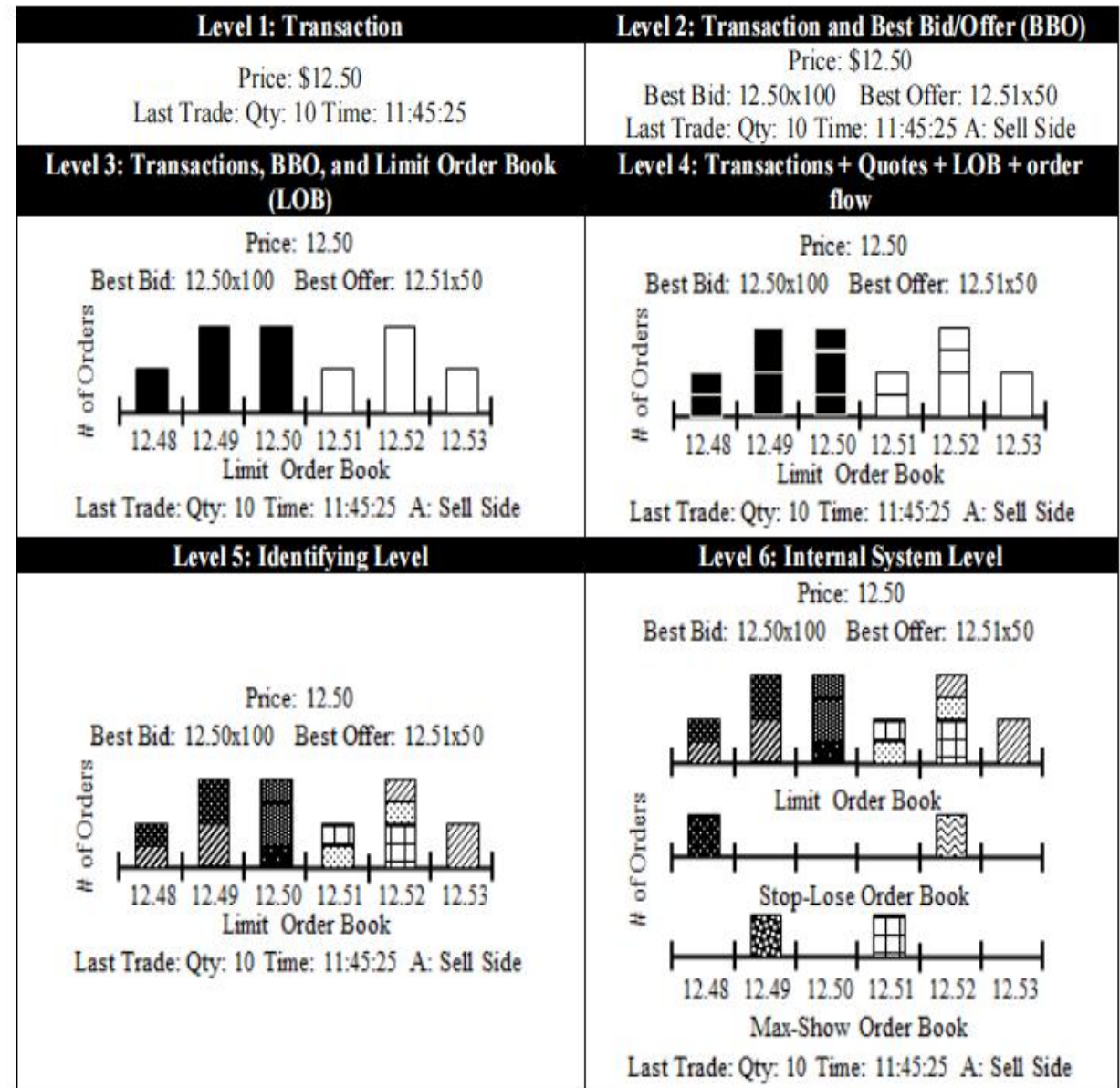
- Available to public
- Shows aggregation of order flow

Level 3 and 4:

- Purchase
- Provide a snap shot view of the limit order book out to some depth (e.g. >10)
- Raw order flow data (Lv 4)

Level 5 and 6:

- Available ONLY to exchanges and regulators
- Identification information
- All events



2. Market Microstructure: Secondary Market Price Discovery

In the secondary market, an asset's price changes only when actual trades happen—when buyers and sellers agree on a transaction price.

- **Price formation = trading activity**

Prices in secondary markets come from the last executed trade (last transaction price).

- **Non-trading assets = no influence**

If large holders keep their assets in cold wallets, vaults, or inactive accounts, that stock/coin/bond doesn't affect the market price until it's put up for sale or bought.

- **Liquidity matters**

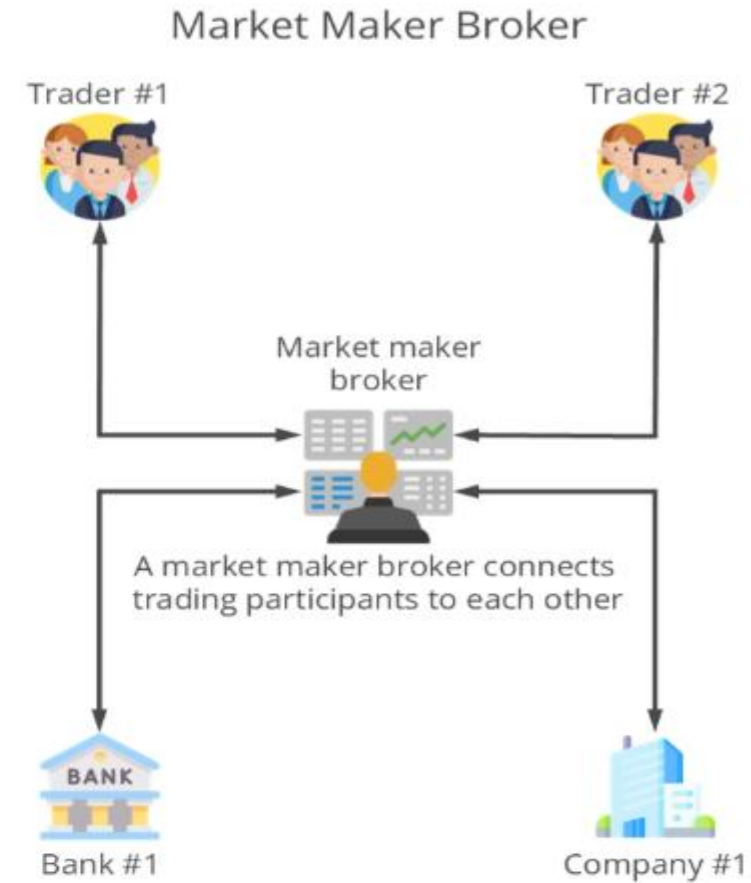
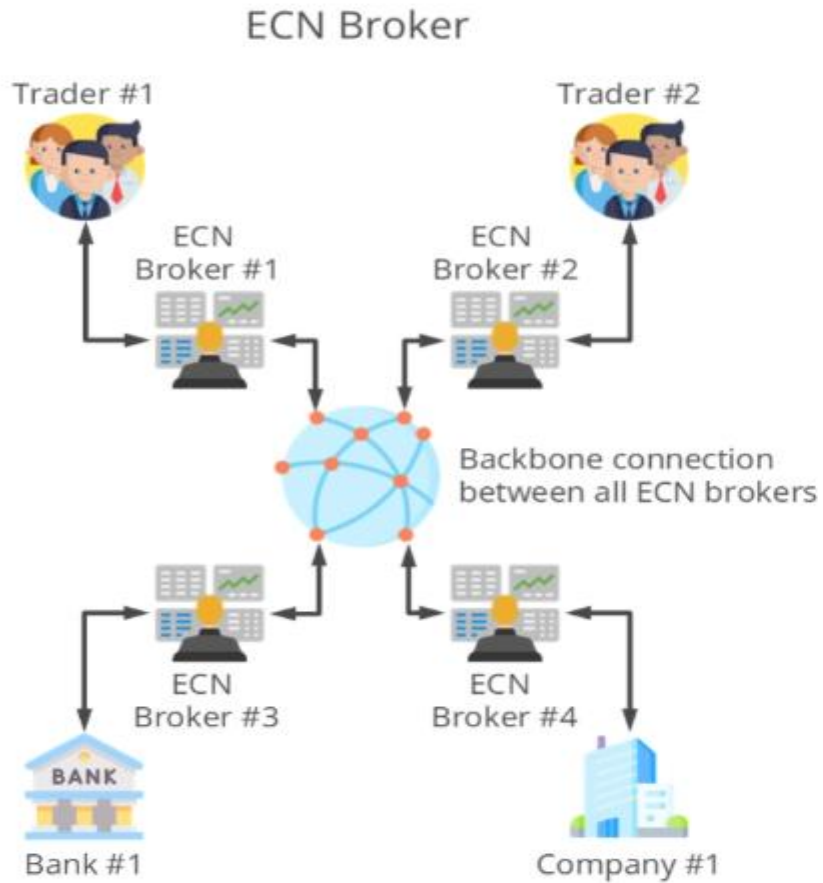
Low-trading assets might have “stale” prices, meaning the quoted price is outdated until a new trade happens.

If a wallet holds 10,000 BTC but doesn't trade them, those coins don't directly affect Bitcoin's market price—only transactions on exchanges do.



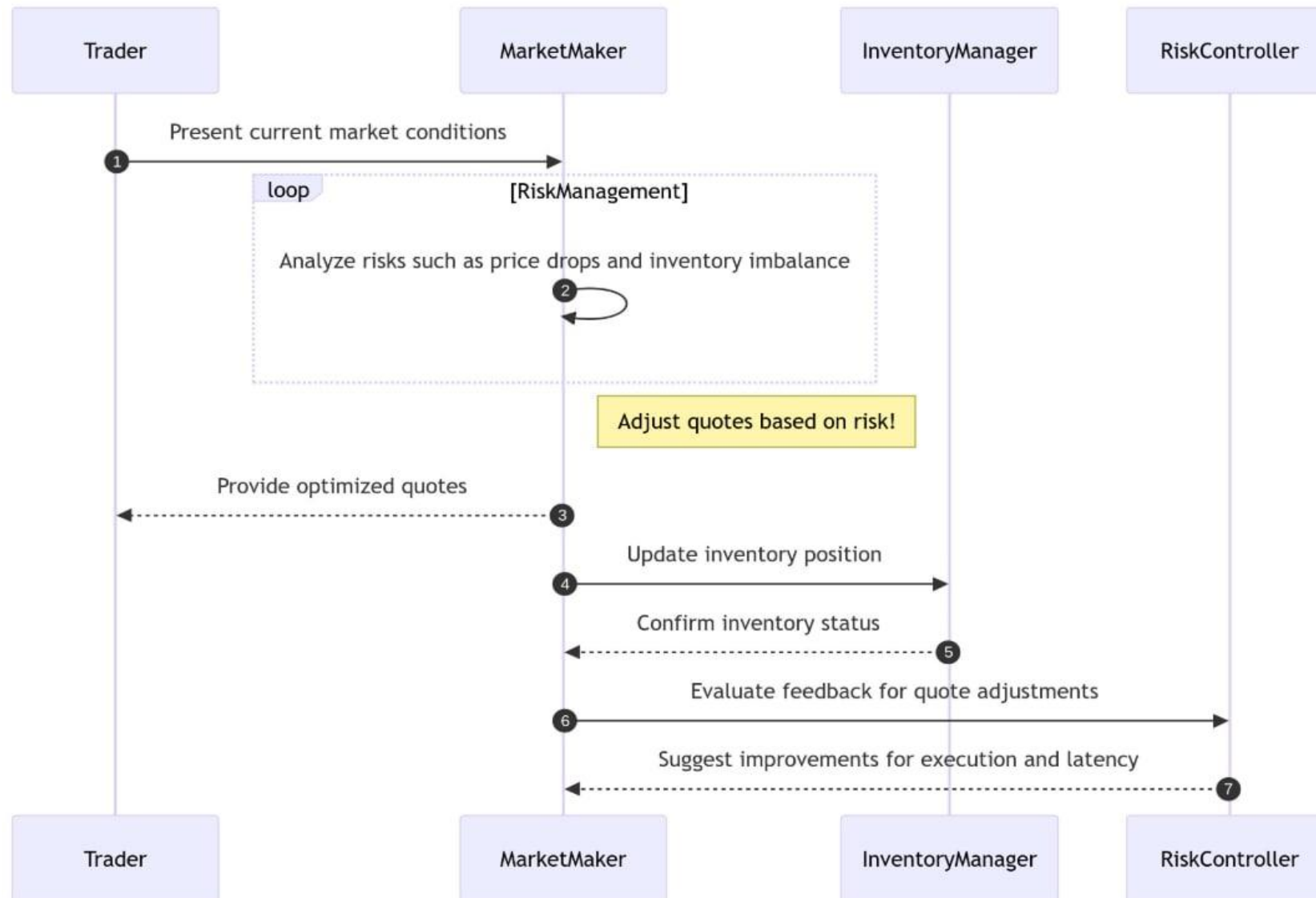
3. Market Making

3. Market Making: Traditional Broker and Market Maker



- A broker only pass the orders for matching a best available price on behalf of the clients.
- A dealer (market maker) acts as the direct counterparty to a client's trade, taking the opposite side of the transaction into its own inventory.

3. Market Making: Traditional Broker and Market Maker



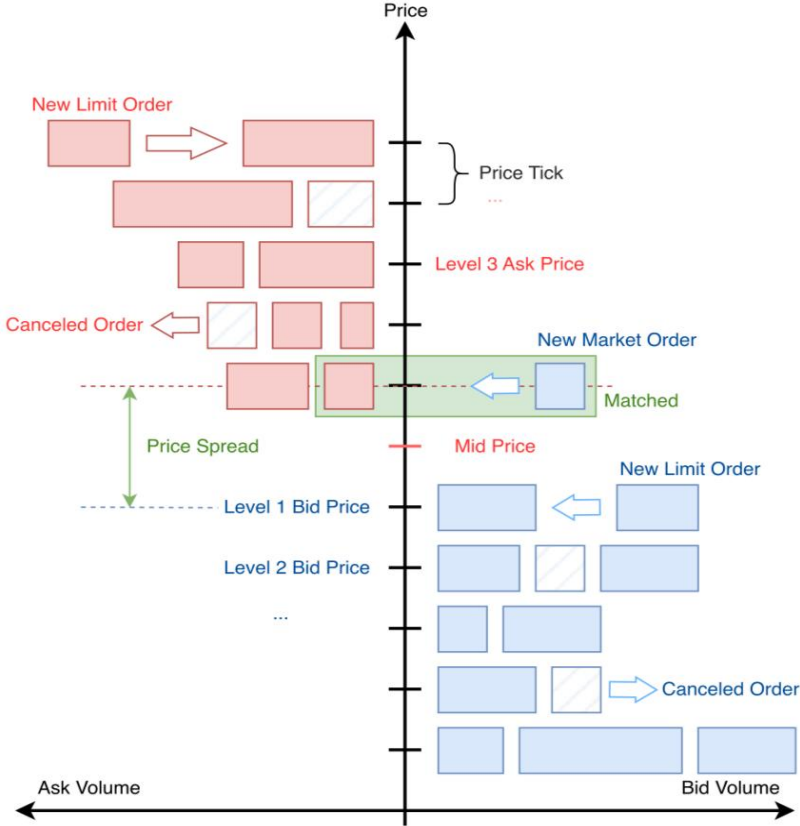
3. Market Making: Traditional Broker and Market Maker

Reality

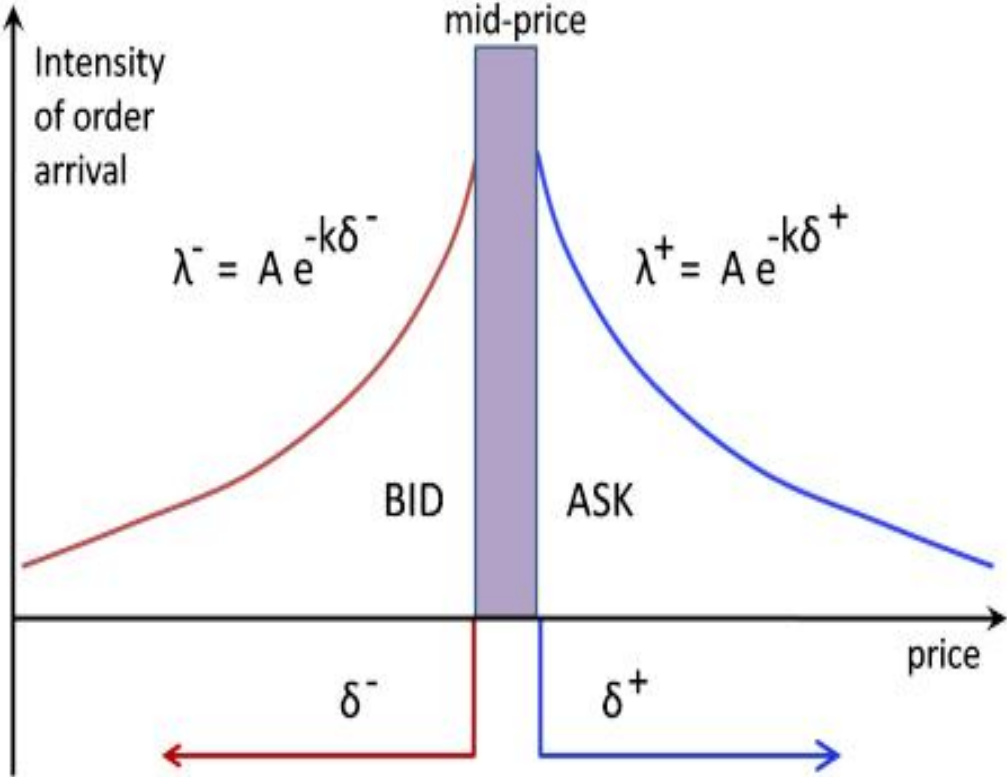


Model

Simplification
Assumption



Source: Muyao Zhong, et al. 2025



Source: Idrees, medium.com

3. Market Making: Avellaneda–Stoikov Model

(Assumption) Price Movement:	Black-Scholes Model:	$dS_t = \mu dt + \sigma dW_t$
	Avellaneda–Stoikov model:	$dS_t = \sigma dW_t$

(Assumption) Intensity of Order Arrival: $\lambda(\delta)$ is Poisson intensity which a limit order will be executed as a function of its distance δ to the mid-price.

$$\lambda(\delta) = Ae^{-k\delta}$$

(Derivation) Reservation Price:
$$r(s, q, t) = s - q\gamma\sigma^2(T - t)$$

(Derivation) Optimal bid-ask spread:
$$\delta^{ask} + \delta^{bid} = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{\kappa}\right)$$

s : current market mid price

q : quantity of assets in inventory of base asset
(positive/negative for long/short positions)

σ : volatility

T : closing time, when the measurement period ends

(usually normalized to 1)

t : current time (As time runs out, the urgency to flatten your position increases, making inventory risk more potent.)

γ : inventory risk aversion parameter.

(A high γ measures means you dislike holding inventory)

κ : orderbook liquidity parameter

3. Market Making: Risk and Limitations of Avellaneda-Stoikov Model

1. Unrealistic model assumptions:

- Prices are modeled as smoothly random with fixed volatility.
- The model assumes random, independent order fills. In reality, orders cluster and influence each other, making fill probabilities less predictable than the model assumes.
- The model assumes your trading doesn't affect the market price.

2. Parameter calibration challenges:

- Pinpointing how order flow responds to your quote distance is complex, requiring analysis of noisy, high-speed data, and these parameters aren't stable.
- The model assumes your tolerance for risk is constant.
- The model uses a basic curve for how fill probability drops with distance. Real market depth is more complex, making the model's spread calculations less precise.

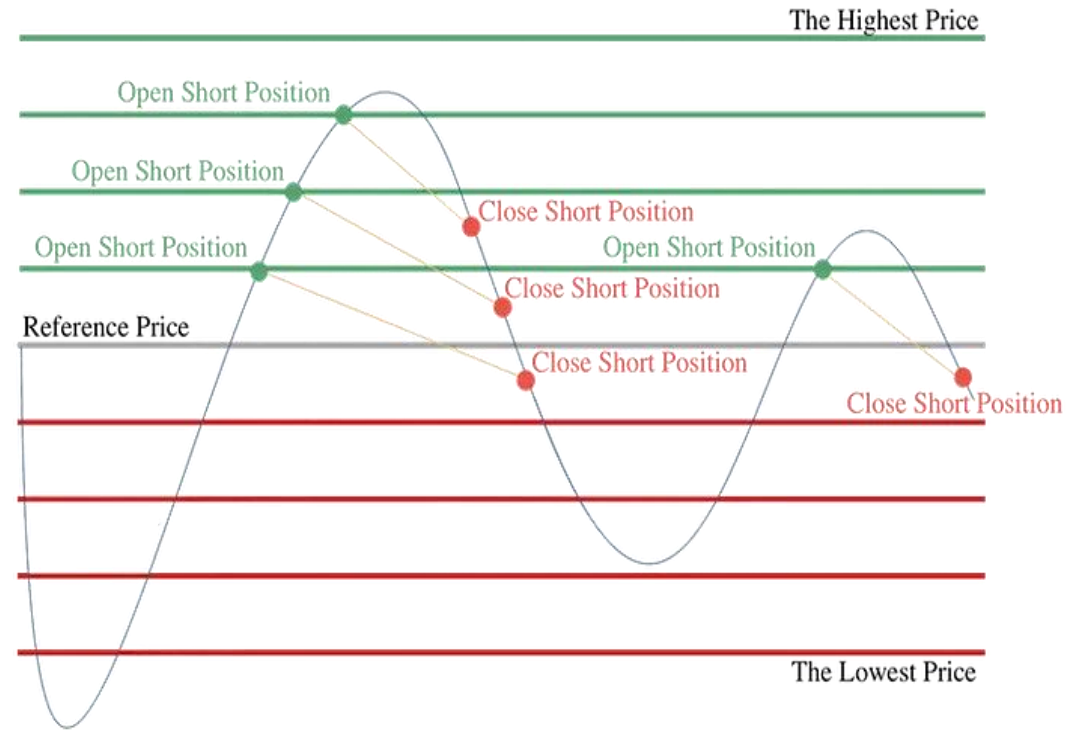
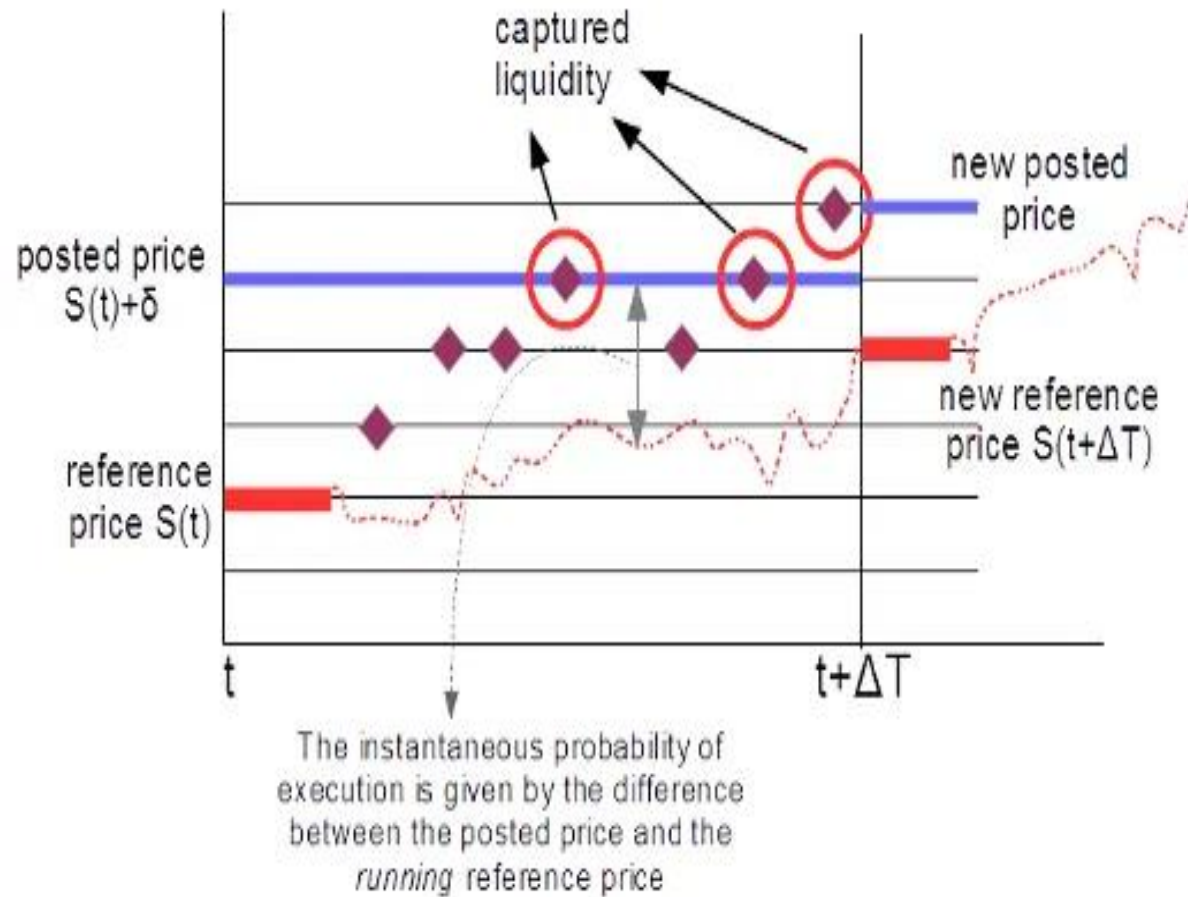
3. Microstructure limitations::

- Markets have minimum price steps and orders are filled in sequence at each price. The continuous model ignores these, missing key factors affecting execution probability and speed.
- The model overlooks the cost of trading, which eats into profits and makes theoretical spreads appear more lucrative than they are.
- The model assumes instantaneous order placement. Delays expose quotes to informed traders who can exploit outdated prices before you can react.

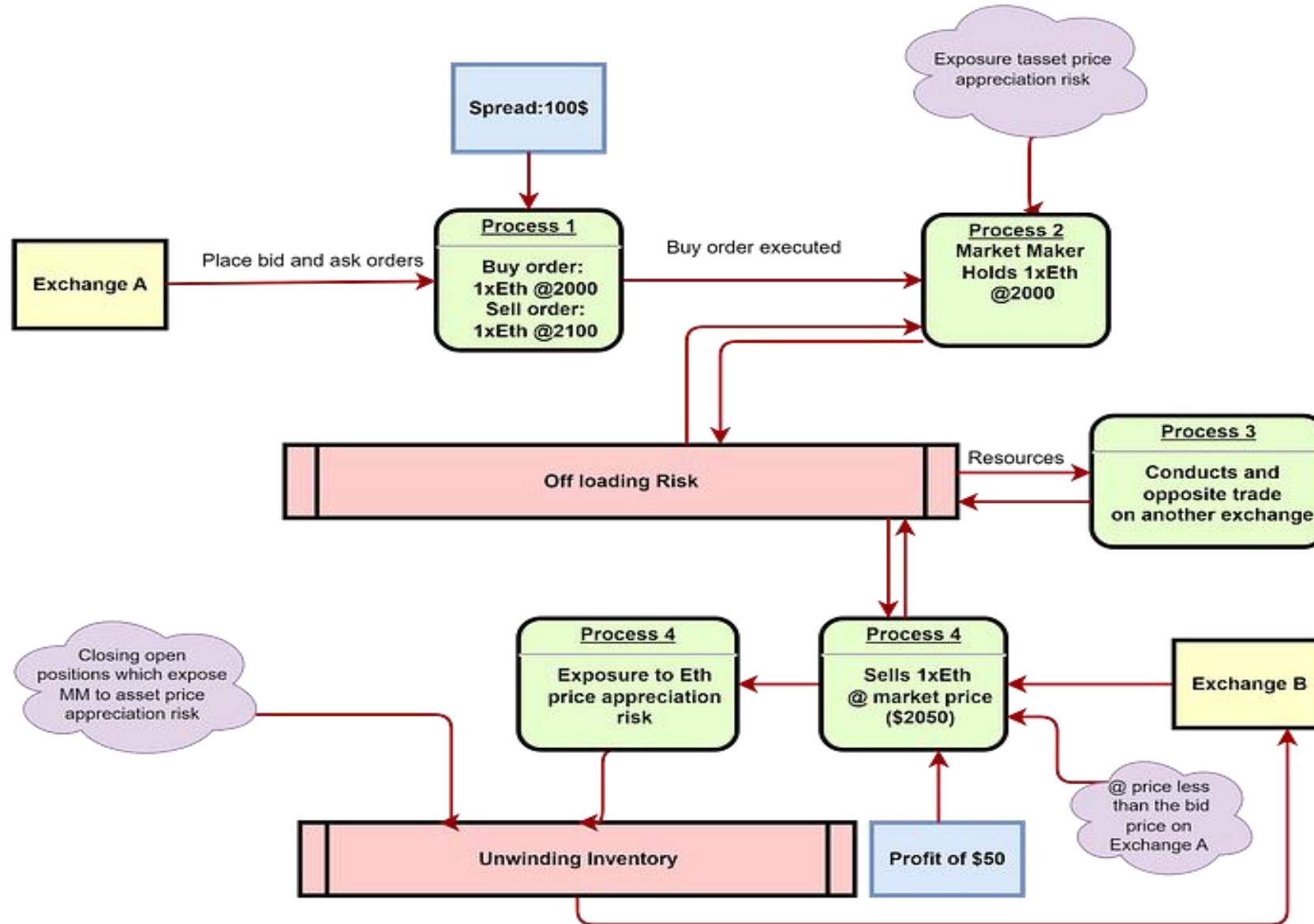
4. Execution Risks:

- Speed, Speed, Speed

3. Market Making: Single-Exchange Market Making



3. Market Making: Cross-Exchange Market Making/Arbitraging



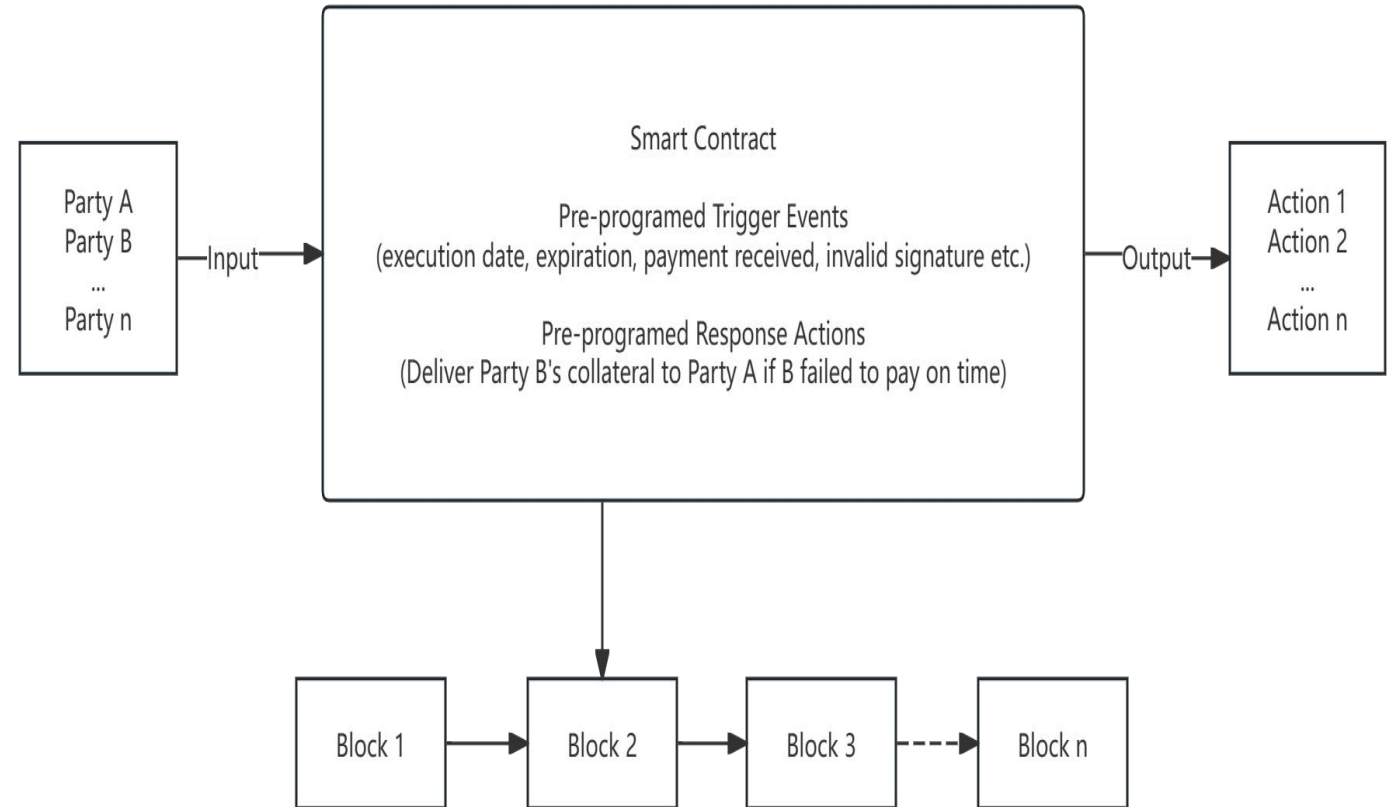
3. Market Making: Smart Contract for Trading

Pros:

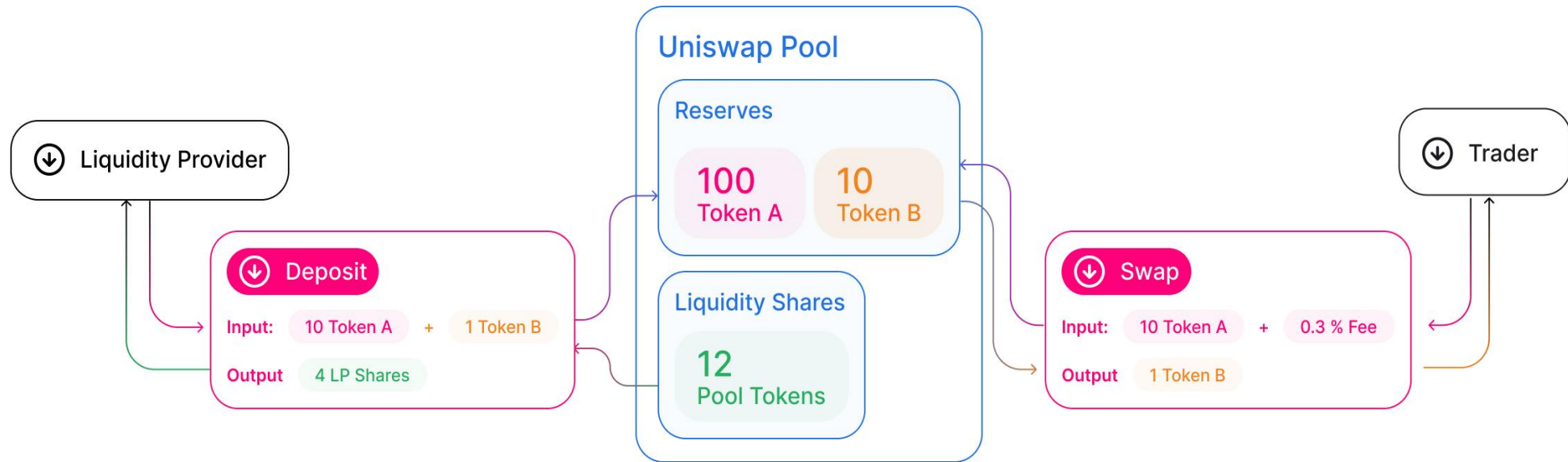
- Executes transactions automatically when conditions are met, reducing manual work and delays.
- Code and rules are visible on the blockchain, so all parties can verify terms, reducing disputes.
- Reduces the need for intermediaries (lawyers, banks, brokers), lowering fees.
- Blockchain encryption makes smart contracts difficult to tamper with once deployed.

Cons:

- Writing error-free smart contracts requires specialized programming knowledge.
- Regulatory frameworks for smart contracts are still evolving; legal enforceability may vary.
- Hard to handle exceptions or changes in real-world circumstances.

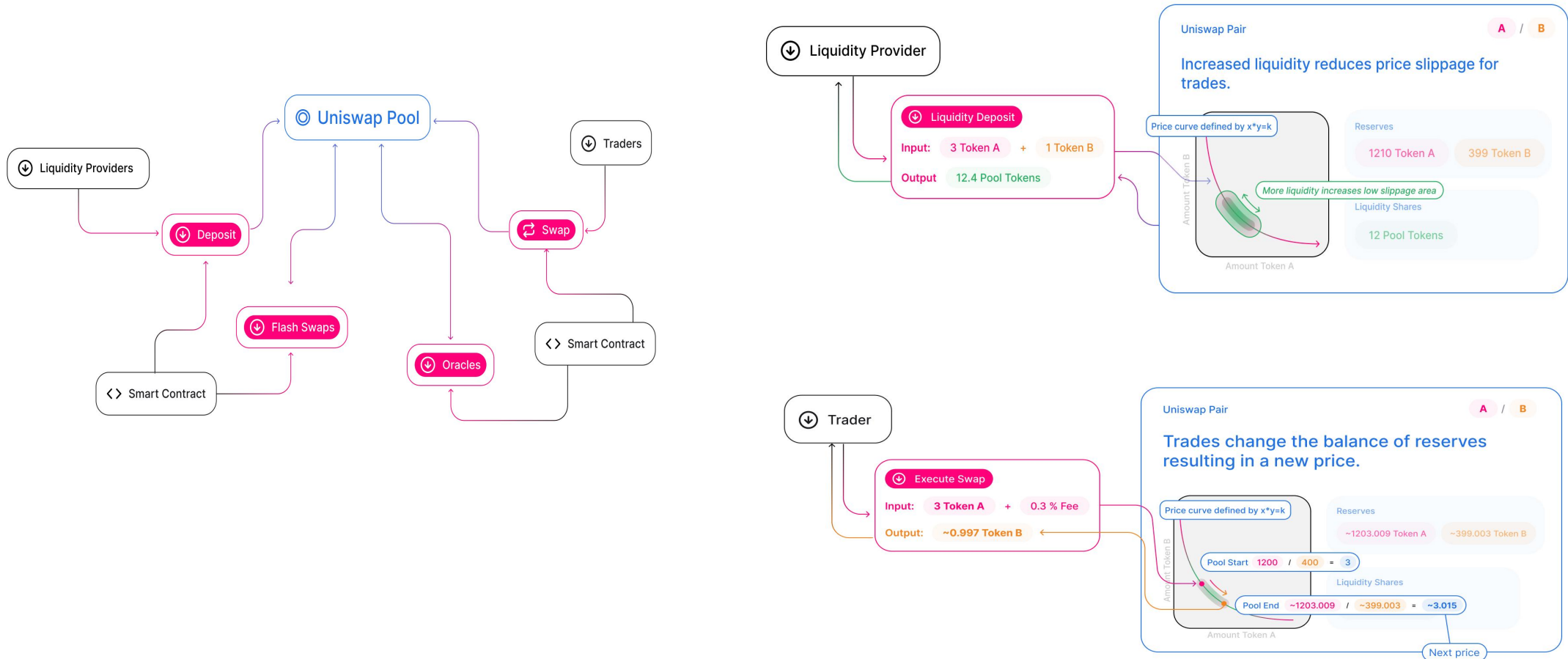


3. Market Making: Decentralized Exchange (Uniswap)



3. Market Making: Decentralized Exchange (Uniswap)

At the core of every AMM is a set of smart contracts. These are self-executing programs deployed to public blockchains like Ethereum, and can power trading logic with no centralized operator required.



3. Market Making: Automated Market Making (AMM)

What is AMM?

- An automated market maker (AMM) is a type of decentralized exchanges that runs on smart contracts, not order books.
- Traditional order books rely on buyers and sellers posting competing offers, and trades only happen when prices match.
- AMMs skip that back-and-forth by using code and math to price trades automatically, enabling fast, 24/7 onchain markets. AMMs can process trillions in trades — no centralized intermediaries required. And any fees are paid transparently to liquidity providers and blockchain networks.

Key Features:

- Anyone can create a new trading pair.
- Prices are automated, not negotiated. AMMs set pricing for assets by following a predictable formula, which ensures transparency and consistency, even during market volatility.
- Open to anyone, anywhere. As decentralized platforms, AMMs are globally accessible and don't require approvals for user activity.
- Earning fees by providing liquidity. In addition to these benefits for traders, there's another opportunity: anyone can add an equal value of two tokens to a liquidity pool and earn a share of the trading fee.

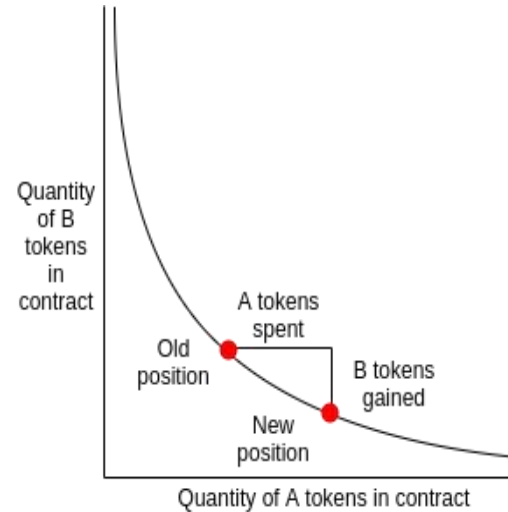
3. Market Making: Automated Market Making (Uniswap's Formula)

$$k=x*y$$

where x is the reserve of Token A, y is the reserve of Token B.

The total value of one token in an AMM liquidity pool always matches the value of the other, keeping the pool balanced at all times.

k is a fixed value that must remain constant after each trade. Liquidity providers set the initial value of k when they create a pool by depositing sums of two tokens in equal value. From that point on, k stays constant during trades.

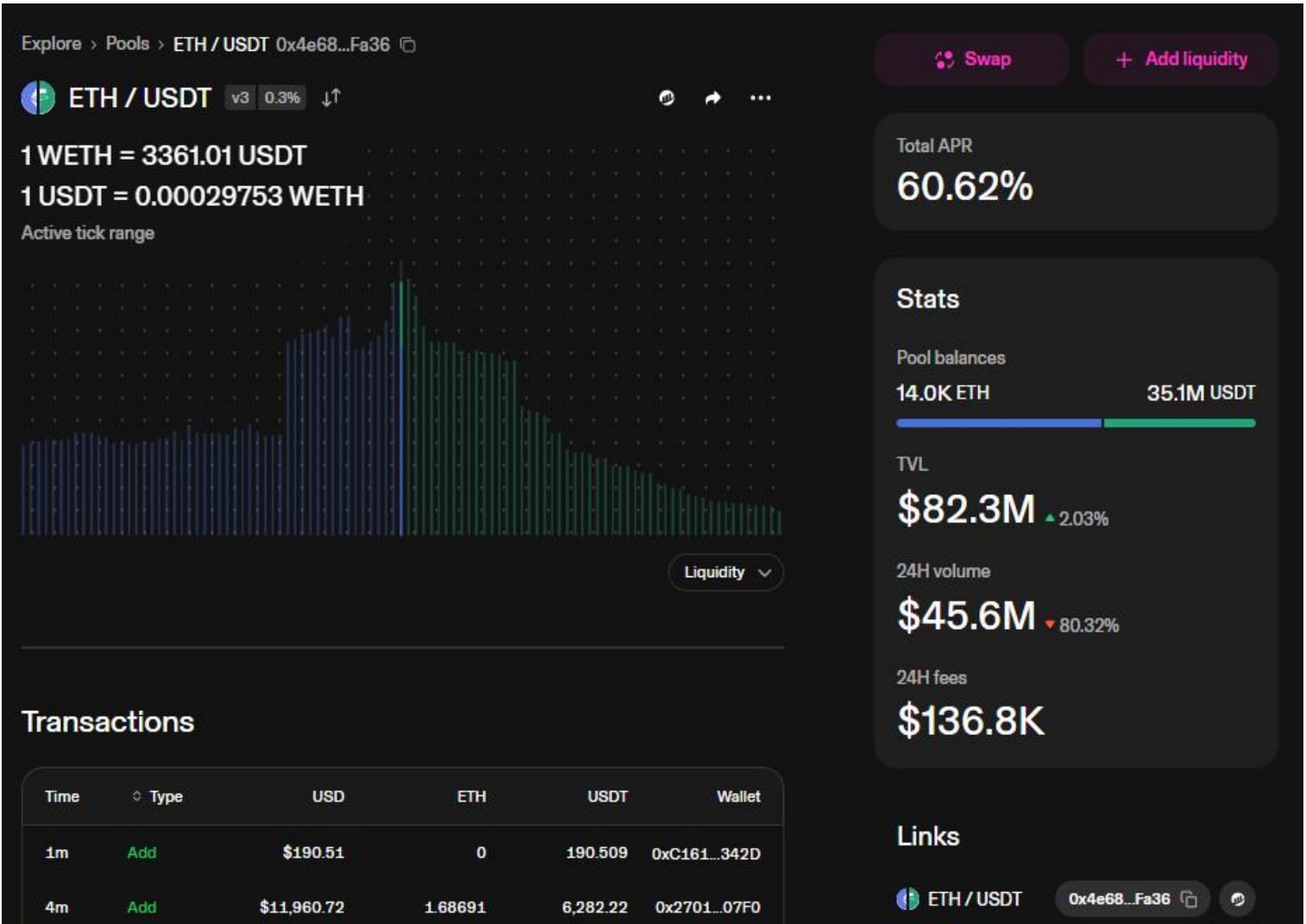


$$\Delta y \leq y - \left(\frac{x \times y}{x + (\Delta x \times 99.7\%)} \right)$$

- x and y represents the quantity of tokens in the liquidity pool before the swap
- Δx represents the amount of token deposited into the AMM
- Δy represents the amount swapped out of the AMM

$$\begin{aligned} \Delta \text{USDC} &= 100 \text{ USDC} - \left(\frac{100 \text{ ETH} \times 100 \text{ USDC}}{124.925 \text{ ETH}} \right) \\ &= 100 \text{ USDC} - \left(\frac{10,000}{124.925} \right) \text{ USDC} \\ &= 100 \text{ USDC} - (80.048 \text{ USDC}) \\ &= 19.952 \text{ USDC} \end{aligned}$$

3. Market Making: Demand/Supply of the Token Pool



source:
<https://app.uniswap.org/explore/pools/ethereum/0x4e68Ccd3E89f51C3074ca5072bbAC773960dFa36?lng=en-US>

3. Market Making: Arbitraging between Traditional MM and Automated MM?

TMM

$$dS_t = \sigma dW_t$$

$$\delta^{ask} + \delta^{bid} = \gamma \sigma^2 (T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{\kappa}\right)$$

AMM

$$\Delta y \leq y - \left(\frac{x \times y}{x + (\Delta x \times 99.7\%)} \right)$$

?

Assignment 1 (Regular Track): Crypto Market Analysis

Lecture 1

- 1.(10) Download data from <https://coinmarketcap.com/historical/> by hand OR through APIs from any crypto exchange you prefer (e.g. coinbase, binance, etc.).
- 2.(10) Check and download data from <https://www.blockchain.com/explorer/charts> , either by hand or through API is fine.
- 3.(30) List the data you think that can be used to predict BTC prices and explain the reasons behind each. Quantitative explanations are preferred.

Lecture 2

- 4.(10) Find and implement a limit order book simulator on Github. Display and explain its outcomes.

Lecture 3

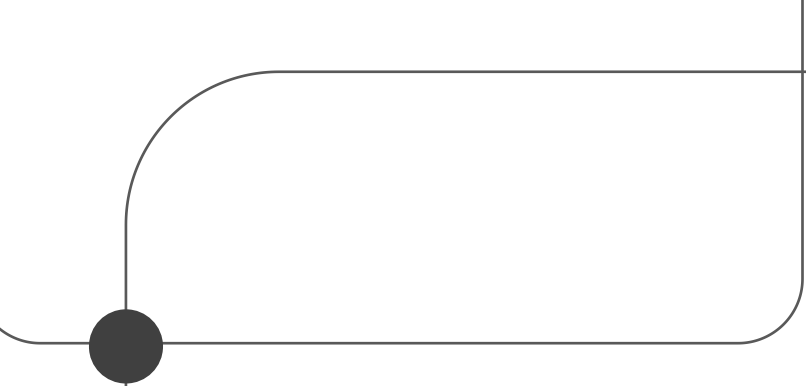
- 5.(10) Locate the crypto asset(s) you want to trade. Describe its statistical features, risk factors, and other information you think is important.
- 6.(30) What kind of strategies do you think could be applied to the asset(s) you selected? Explain how it is related to your strategy.

Assignment 1 (Express Track): Crypto Market Analysis

Challenge 1(100): Implement the factor model from the paper: Liu, Y., Tsyvinski, A., & Wu, X. (2022). Common risk factors in cryptocurrency. *The Journal of Finance*, 77(2), 1133-1177.

The paper's factor model excludes blockchain-specific variables because it is designed as a general asset-pricing framework rather than a BTC-focused predictor. Can you combine both blockchain factors and price factors together for BTC and ETH? This problem serves as preparation for designing market-neutral strategies.

Challenge 2(100): Write a simulator for automated market maker (AMM) from Uniswap. Compare the price/liquidity generated from the traditional simulator against the AMM. The purpose of this problem is to prepare for developing a cross-exchange arbitrage strategy.



Thank You

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